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3. Consider a simple pendulum of length $l = 0.9\ m$ which is properly placed on a trolley rolling down on a inclined plane which is at $\theta = 45°$ with the horizontal. Assuming that the inclined plane is frictionless, calculate the time period of oscillation of the simple pendulum.

Answer: 0.86 s

4. A piece of wood of mass m is floating erect in a liquid whose density is ρ. If it is slightly pressed down and released, then executes simple harmonic motion. Show that its time period of oscillation is $T = 2\pi\sqrt{\frac{m}{Ag\rho}}$

5. Consider two simple harmonic motion along x and y-axis having same frequencies but different amplitudes as $x = A \sin(\omega t + \varphi)$ (along x axis) and $y = B \sin \omega t$ (along y axis). Then show that

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} - \frac{2xy}{AB}\cos\varphi = \sin^2\varphi$$

and also discuss the special cases when

a. $\varphi = 0$ b. $\varphi = \pi$ c. $\varphi = \frac{\pi}{2}$

d. $\varphi = \frac{\pi}{2}$ and $A = B$ (e) $\varphi = \frac{\pi}{4}$

Note: *when a particle is subjected to two simple harmonic motion at right angle to each other the particle may move along different paths. Such paths are called* ***Lissajous figures****.*

Answer :

a. $y = \frac{B}{A}x$, equation is a straight line passing through origin with positive slope.

b. $y = -\frac{B}{A}x$ equation is a straight line passing through origin with negative slope.

c. $\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$, equation is an ellipse whose center is origin.

d. $x^2 + y^2 = A^2$, equation is a circle whose center is origin .

e. $\frac{x^2}{A^2} + \frac{y^2}{B^2} - \frac{2xy}{AB}\frac{1}{\sqrt{2}} = \frac{1}{2}$, equation is an ellipse (oblique ellipse which means tilted ellipse)

6. Show that for a particle executing simple harmonic motion

a. the average value of kinetic energy is equal to the average value of potential energy.

b. average potential energy = average kinetic energy = $\frac{1}{2}$(total energy)

Hint : *average kinetic energy* = $\langle$*kinetic energy*$\rangle = \frac{1}{T}\int_0^T \left(Kinetic\,energy\right)dt$

and

average Potential energy = $\langle$*Potential energy*$\rangle = \frac{1}{T}\int_0^T \left(Potential\ energy\right)dt$

7. Compute the time period for the following system if the block of mass m is slightly displaced vertically down from its equilibrium position and then released. Assume that the pulley is light and smooth, strings and springs are light.

